

Counters in Digital Logic

A **Counter** is a device which stores (and sometimes displays) the number of times a particular event or process has occurred, often in relationship to a clock signal. Counters are used in digital electronics for counting purpose, they can count specific event happening in the circuit. For example, in UP counter a counter increases count for every rising edge of clock. Not only counting, a counter can follow the certain sequence based on our design like any random sequence 0,1,3,2.... They can also be designed with the help of flip flops. They are used as frequency dividers where the frequency of given pulse waveform is divided. Counters are sequential circuit that count the number of pulses can be either in binary code or BCD form. The main properties of a counter are timing, sequencing, and counting. Counter works in two modes
Up counter

Down counter

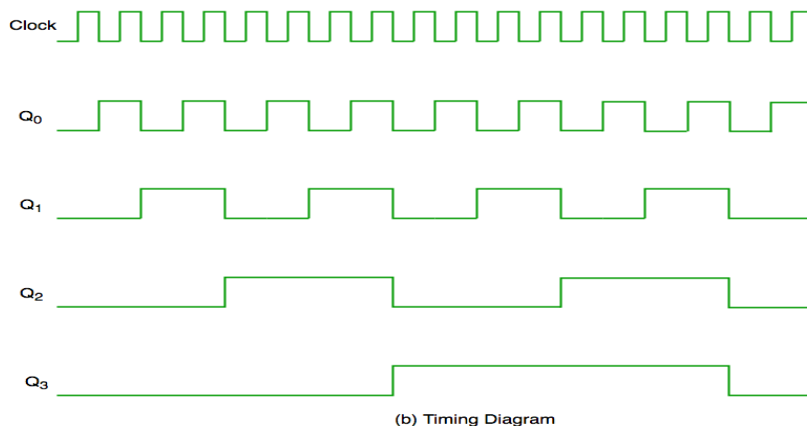
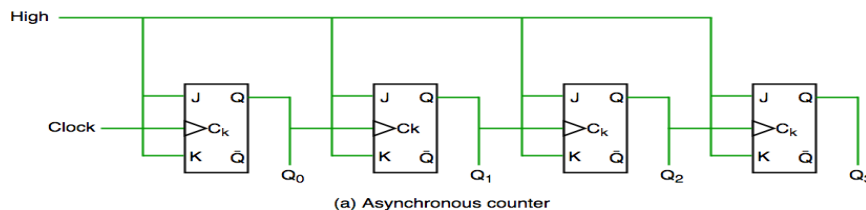
Counter Classification

Counters are broadly divided into two categories

1. Asynchronous counter
2. Synchronous counter

1. Asynchronous Counter

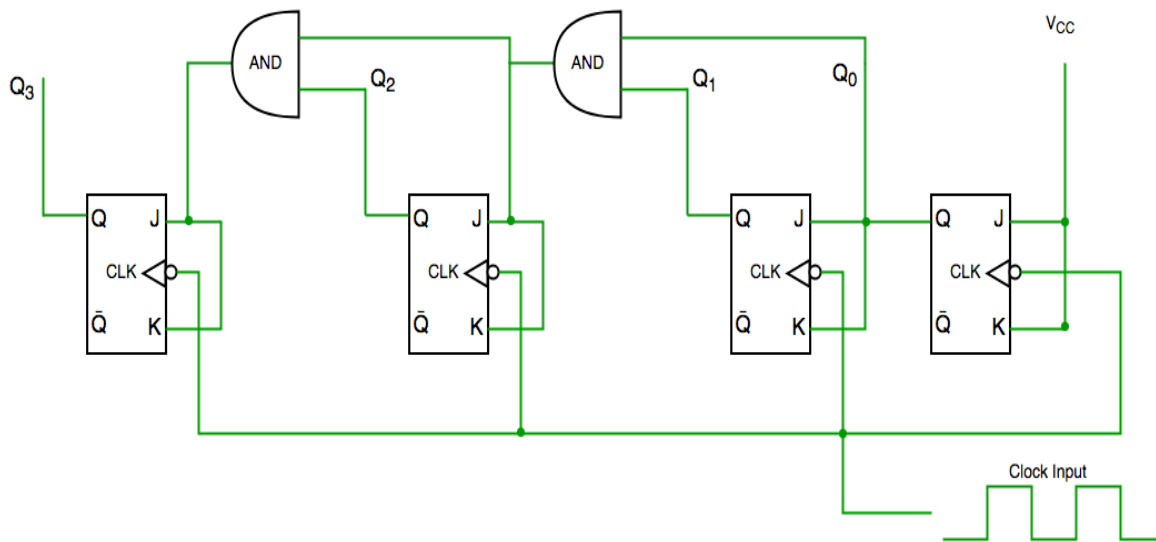
In asynchronous counter we don't use universal clock, only first flip flop is driven by main clock and the clock input of rest of the following flip flop is driven by output of previous flip flops. We can understand it by following diagram-



It is evident from timing diagram that Q0 is changing as soon as the rising edge of clock pulse is encountered, Q1 is changing when rising edge of Q0 is encountered (because Q0 is like clock pulse for second flip flop) and so on. In this way ripples are generated through Q0, Q1, Q2, Q3 hence it is also called **RIPPLE counter and serial counter**. A ripple counter is a cascaded arrangement of flip flops where the output of one flip flop drives the clock input of the following flip flop.

2. Synchronous Counter

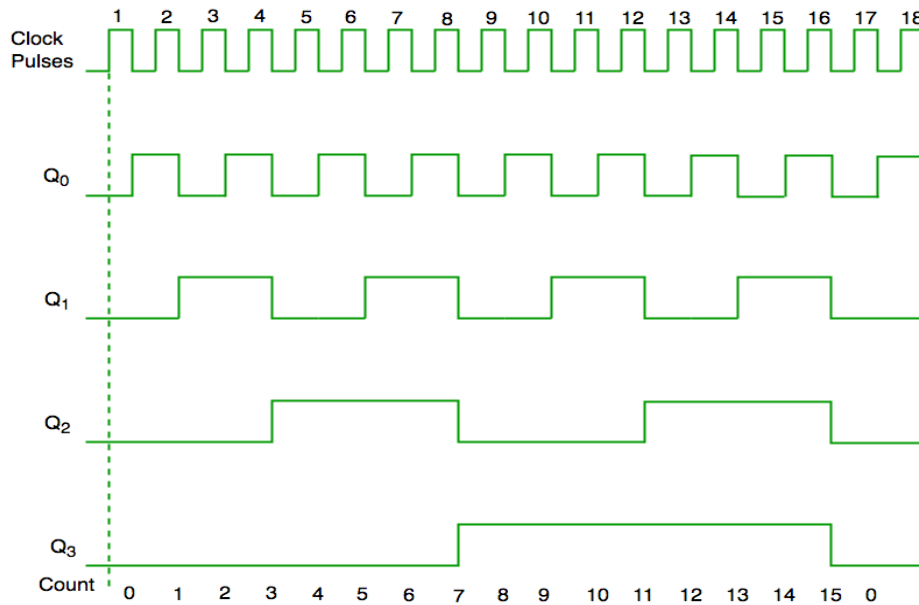
Unlike the asynchronous counter, synchronous counter has one global clock which drives each flip flop so output changes in parallel. The one advantage of synchronous counter over asynchronous counter is, it can operate on higher frequency than asynchronous counter as it does not have cumulative delay because of same clock is given to each flip flop. It is also called as parallel counter.



nchronous

counter

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circuit



Timing diagram synchronous counter

From circuit diagram we see that Q_0 bit gives response to each falling edge of clock while Q_1 is dependent on Q_0 , Q_2 is dependent on Q_1 and Q_0 , Q_3 is dependent on Q_2 , Q_1 and Q_0 .

Synchronous vs. Asynchronous Counter

Counters are essential components in digital electronics used for counting events or time.

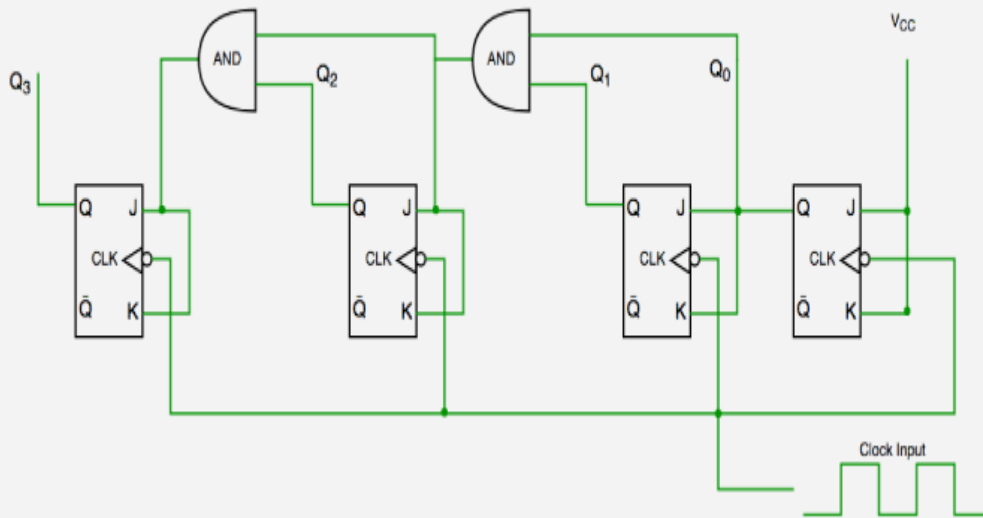
- They are widely used in digital systems for tasks like frequency division, event counting, and time measurement.
- Their design and operation play a crucial role in determining the accuracy and efficiency of digital circuits.

Synchronous Counter

A Synchronous Counter is a counter where all flip-flops are triggered simultaneously by a single common clock signal, ensuring synchronized state changes.

- A universal clock controls all flip-flops at once.
- All flip-flops change state together for precise timing.

Synchronous Counter



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nchronous Counter

Advantages of Synchronous Counters:

- **Faster Operation:** All flip-flops trigger simultaneously for quicker response.
- **Precise Timing:** Synchronized operation reduces timing errors.
- **Low Propagation Delay:** No ripple effect between flip-flops.
- **High Reliability:** Stable and error-resistant performance.
- **Ideal for High-Speed Use:** Suitable for fast timers and processors.
- **Simplified Control:** Single clock source makes timing easier to manage.

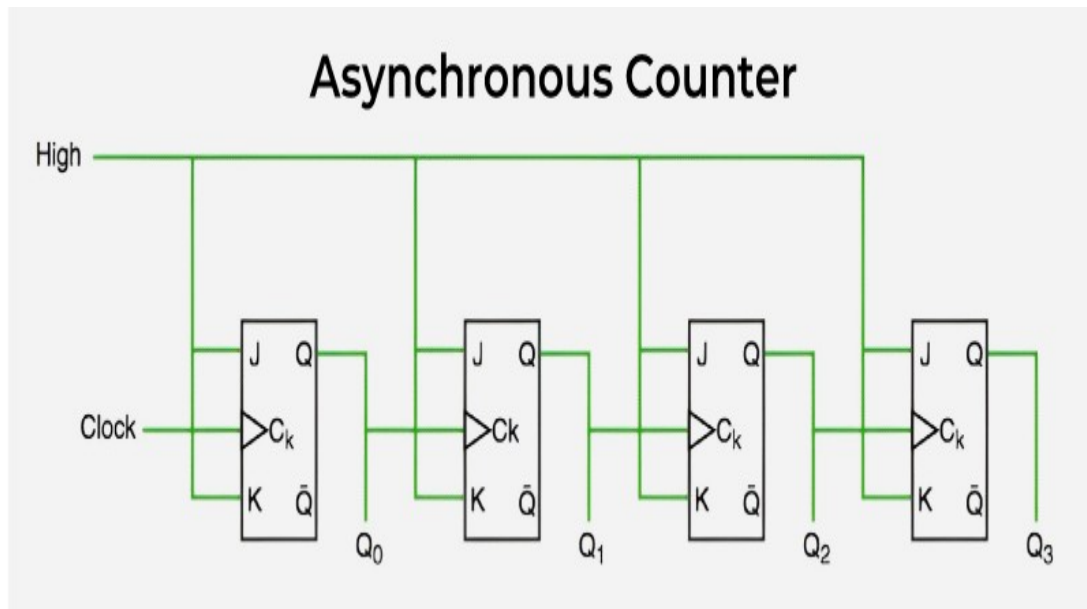
Disadvantages of Synchronous Counters:

- **Complex Design:** Requires additional control logic and synchronization circuits.
- **Higher Power Consumption:** All flip-flops switching simultaneously increases power usage.
- **More Components Needed:** Needs extra hardware like clock drivers and buffers.

Asynchronous Counter

An Asynchronous Counter, or Ripple Counter, is a counter where only the first flip-flop receives the main clock signal, and each subsequent flip-flop is triggered by the output of the previous one.

- Flip-flops change state one after another, causing a ripple effect.
- Only the first flip-flop gets the external clock; others rely on preceding outputs.



Advantages of Asynchronous Counters:

- **Simple Design:** Each flip-flop triggers the next, no common clock needed.
- **Lower Power Consumption:** Flip-flops change one at a time, saving power.
- **Cost-Effective:** Fewer components make it cheaper to build.
- **Good for Low-Speed Use:** Suitable for basic timers and event counters.
- **Flexible:** Works well when precise synchronization isn't required.

Disadvantages of Asynchronous Counters:

- **Slower Operation:** Sequential triggering causes delays.
- **Timing Issues:** Ripple effect can lead to errors at high speeds.
- **Less Accurate:** Propagation delays reduce precision.
- **Limited Speed:** Not suitable for fast applications.
- **Reduced Reliability:** More prone to glitches as flip-flops increase.

Synchronous vs. Asynchronous Counter

Synchronous Counter	Asynchronous Counter
In synchronous counter we use a universal clock that is common to all flip flops through out the circuit.	In asynchronous counter main clock is only applied to the first flip flop and then for rest of flip flops the output of previous flip flop is taken as a clock.
Synchronous Counter is faster in operation as compared to	Asynchronous Counter is slower as compared to

Synchronous Counter	Asynchronous Counter
Asynchronous Counter.	synchronous counter in operation.
Synchronous Counter does not produce any decoding errors.	Asynchronous Counter produces decoding error.
Synchronous Counter is also called Parallel Counter.	Asynchronous Counter is also called Serial Counter.
Synchronous Counter designing as well implementation are complex due to increasing the number of states.	Asynchronous Counter designing as well as implementation is very easy.
Synchronous Counter will operate in any desired count sequence.	Asynchronous Counter will operate only in fixed count sequence (UP/DOWN).
Synchronous Counter examples are: <u>Ring counter</u> , <u>Johnson counter</u> .	Asynchronous Counter examples are: <u>Ripple</u> UP counter, Ripple DOWN counter.
In synchronous counter, propagation delay is less.	In asynchronous counter, there is high propagation delay.